Presentation on Compressed Earth Blocks (CEB)

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Background of Habitat for Humanity Nepal

Mission

Habitat Nepal brings people together to build homes, communities & hope.

- Focus on Shelter
- Advocate for affordable technology
- Promote dignity & hope
- Support sustainable & transformational development

Project location & partner organization

Habitat Nepal is committed to building homes & hope in both disaster affected (Kavre & Nuwakot) & Non affected areas (Eastern terai & Western terai)

Avari Foundation is a partner organization helping on promotion of CEB in Habitat project location.

Experience with CEB pre-earthquake

Photo 1: HFHN prepared adobe bricks

Adobe bricks

- Most selected construction materials by house owner in our project areas.

HFHN support

- Grant through MFI
- Training to improve quality
- National & international volunteer support as labour.

CEB

Photo 2: Making CEB with locally available mud

CEB

- Mainly focus in terai and DR area in Panchkhal municipality, Kavre.
- Good quality of mud available in these areas.
- Mortar for CEB laying & plaster = mix for brick

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Photo3: Laying bricks in wall in super structure

Photo 4: Drying process of CEB

Activities for Promotion of CEB

Photo 5: CEB preparing machine

Construction equipment library

- Mechanical compressor machine
- Available 1-4 compressor machine within each project location.
- House owner can take this machine to make CEB for their house from this Library.

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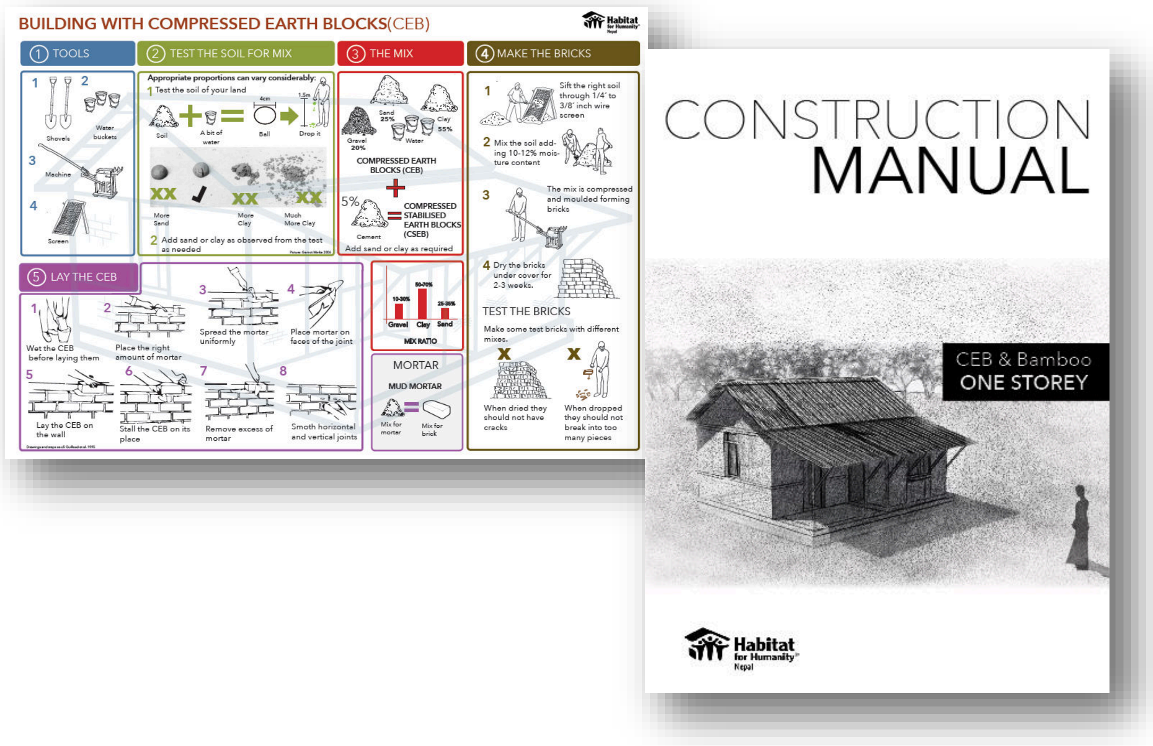
Photo 6: Poster & manual



Construction manual and poster on CEB

- Manual for one storey non incremental and one storey with attic
- Poster on CEB making process

Contd..

Photo 7 : Design for model house

Model house

- In Panchkhal, through “on the Job training”.
- House with stone, CEB & Bamboo.
- Through 30 unskilled mason

Contd..

Photo 8: Participants in CEB making training, panchkhal

Market development & livelihood training

- 20 participants (10 skilled labour & 10 unskilled labour (in Panchkhal)
- 30 participants in Dang.
- Planning to establish one to two CEB manufacturing business enterprises in each project location through intensive business development training.

Approval process after earthquake

Photo 9 : 3-D view of design for submitted for approval

Key activities for approval process

- Design of house with CEB, Stone & Bamboo.
    - Plinth area = 441 sq. ft
    - Two rooms of 12’4” x8’9”
    - Kitchen & Verandha of same size = 5’9” x 9’10
    - Reinforce vertically with rebars and horizontally through concrete banding at floor, sill and lintel levels
    - Conner & T stitch in every 2’
- Structural Design for technically safe
- Submission of drawing to DUDBC for approval

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Photo 10: section of our design

Conclusion

- Technically feasible
- Financially low cost house
- Socially acceptable

Thank You